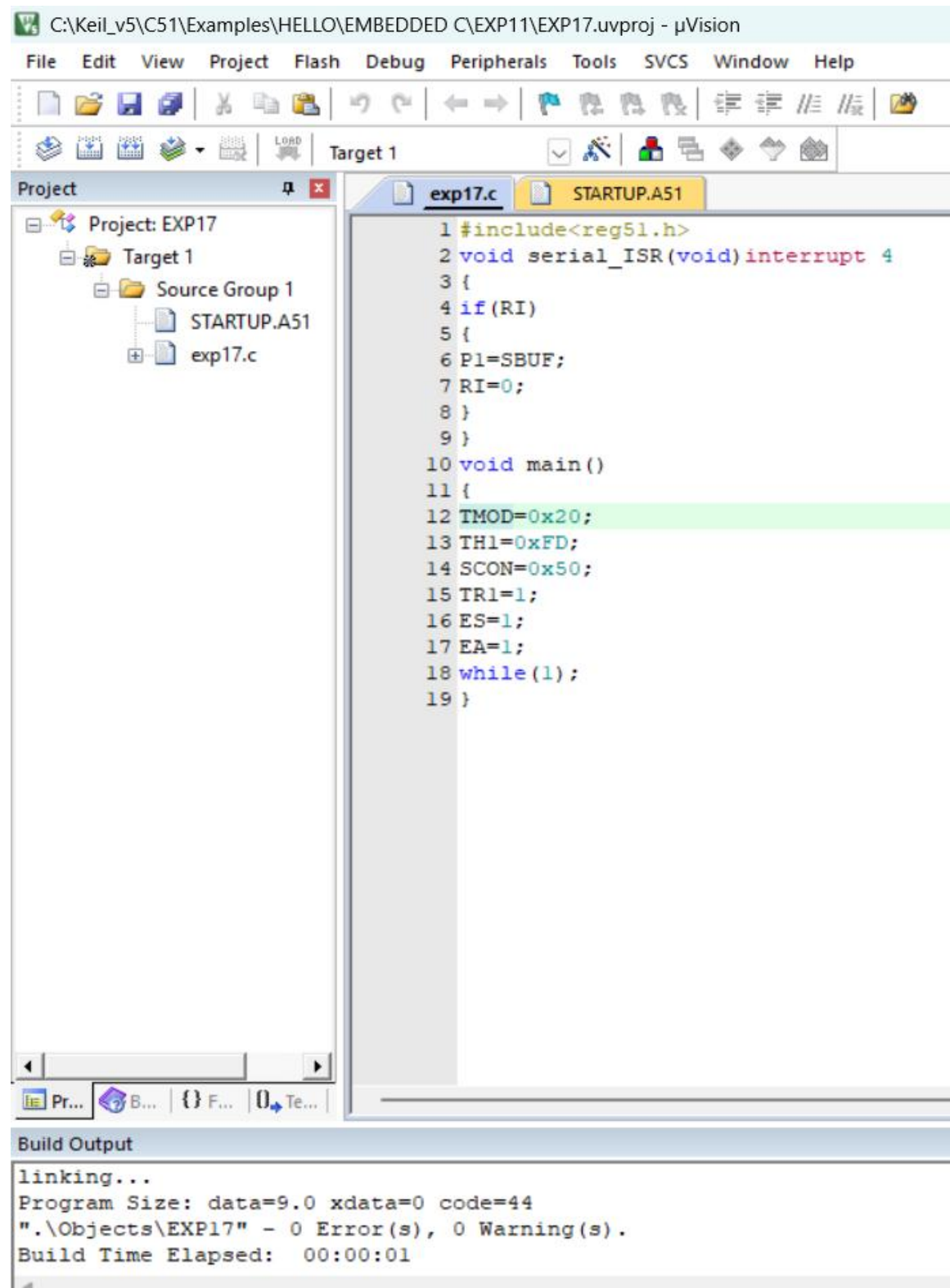


Exp No: 17 UART INTERFACING USING INTERRUPT IN KEIL
DATE: 18/08/2026
PROGRAM:



C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\EXP17.uvproj - µVision

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Project: EXP17

- Target 1
 - Source Group 1
 - STARTUP.A51
 - exp17.c

```
1 #include<reg51.h>
2 void serial_ISR(void) interrupt 4
3 {
4     if (RI)
5     {
6         P1=SBUF;
7         RI=0;
8     }
9 }
10 void main()
11 {
12     TMOD=0x20;
13     TH1=0xFD;
14     SCON=0x50;
15     TR1=1;
16     ES=1;
17     EA=1;
18     while (1);
19 }
```

Build Output

linking...

Program Size: data=9.0 xdata=0 code=44

".\Objects\EXP17" - 0 Error(s), 0 Warning(s).

Build Time Elapsed: 00:00:01

C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\EXP17.uvproj - µVision

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Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC	0x0800
status	389
sec	0.000194...
psw	0x00

Disassembly

```

10: void main()
11: {
12: TMOD=0x20;
13: C:0x0800 758920 MOV TMOD(0x89),#0x20

```

exp17.c STARTUP.A51

```

1 #include<reg51.h>
2 void serial_ISR(void) interrupt 4
3 {
4 if(RI)
5 {
6 P1=SBUF;
7 RI=0;
8 }
9 }
10 void main()
11 {
12 TMOD=0x20;
13 TH1=0xFD;
14 SCON=0x50;
15 TR1=1;
16 ES=1;
17 EA=1;
18 while(1);
19 }

```

Parallel Port 1

Port 1

P1: 0x41 7 Bits 0

Pins: 0x41

Command

Running with Code Size Limit: 2K

Load "C:\\Keil_v5\\C51\\Examples\\HELLO\\EMBEDDED C\\EXP11\\Objects\\EXP17"

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

UART #1

Simulation

C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\EXP17.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC	0x0800
status	389
sec	0.000194...
psw	0x00

Disassembly

```

10: void main()
11: {
12: TMOD=0x20;
13: C:0x0800 758920 MOV TMOD(0x89),#0x20

```

exp17.c STARTUP.A51

```

1 #include<reg51.h>
2 void serial_ISR(void) interrupt 4
3 {
4 if(RI)
5 {
6 P1=SBUF;
7 RI=0;
8 }
9 }
10 void main()
11 {
12 TMOD=0x20;
13 TH1=0xFD;
14 SCON=0x50;
15 TR1=1;
16 ES=1;
17 EA=1;
18 while(1);
19 }

```

Parallel Port 1

Port 1

P1: 0x31 7 Bits 0

Pins: 0x31

Command

Running with Code Size Limit: 2K

Load "C:\\Keil_v5\\C51\\Examples\\HELLO\\EMBEDDED C\\EXP11\\Objects\\EXP17"

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

UART #1

Simulation

C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\EXP17.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC	0x0800
status	389
sec	0.000194...
psw	0x00

Disassembly

```
10: void main()
11: {
12: TMOD=0x20;
C:0x0800 758920 MOV TMOD(0x69),#0x20
```

exp17.c STARTUP.AS1

```
1 #include<reg51.h>
2 void serial_ISR(void) interrupt 4
3 {
4 if (RI)
5 {
6 P1=SBUF;
7 RI=0;
8 }
9 }
10 void main()
11 {
12 TMOD=0x20;
13 TH1=0xFD;
14 SCON=0x50;
15 TR1=1;
16 ES=1;
17 EA=1;
18 while(1);
19 }
```

Parallel Port 1

Port 1

P1: 0x61 7 Bits 0

Pins: 0x61

Command

Running with Code Size Limit: 2K

Load "C:\\Keil_v5\\C51\\Examples\\HELLO\\EMBEDDED C\\EXP11\\Objects\\EXP17"

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

UART #1